

Praktikum Design of Wind Farms

Lecture 1:
Background and Motivations

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Syllabus

No.	Week	Date (Wed)	9.30–10.15	10.30–11.15	11.30–12.15
1	14	15/04/26	Overview of Course Lecture 1: Background and Motivations	Lecture 2: Development of a Wind Farm Project	Lecture 3: Wake Behavior, Wind Farm Control Assignment 1: Basics and Recapitulation
2	15	22/04/26	Lecture 4: Wake Models, FLORIS	Assignment 1: Solution Assignment 2: Jensen's Wake Model and Wake Superposition in Wind Farms	Tutorial 1: FLORIS Installation and Setup
3	16	29/04/26	Lecture 5_1: LandBOSSE	Tutorial 2: LANDBOSSE Installation and Setup	Assignment 2: Solution Assignment 3: Wind Farm Analysis Using FLORIS
5	17	06/05/26	Lecture 5_2: Financial Analysis	Assignment 3: Solution	Assignment 4: Wind Farm Cost Analysis Using FLORIS + LandBOSSE)
6	18	13/05/26	Lecture 6: Layout Optimization	Assignment 4: Solution	Assignment 5: Layout Optimization
7	19	20/05/26	Assignment 5: Solution		Final group assignment: Introduction, Finalizing Groups, Q&A
8	20	27/05/26	Final group assignment: Design Process, Q&A, Writing Report		
9	21	03/06/26	No Class		
10	22	10/06/26	Final group assignment: Design Process, Q&A, Writing Report		
11	28	08/07/26	No Class Deadline: Electronic report submission until Tuesday, 07/07/2026, 23:59h		
12	29	15/07/26	Final presentation – 8:30h (Room 2701M)		



Agenda

- **Goal**
- **Background**
- **Motivations**
- **Wind Energy Statistics**
- **Issues Affecting Growth**
- **Wind Energy Impacts**



**What if onshore/offshore
wind farm layouts were
designed by single
discipline experts?**



Ocean

We have to install
these wind turbines
offshore, but where?



To show the value of using systems engineering for optimising the design of offshore wind farms, I speculate about how absurd (and sub-optimal) they would look if they were designed by experts from single disciplines.

Land



2 The energy yield team

Windier

Less windy

To increase the AEP (annual energy production), turbines are placed at the windiest region and spaced apart as far as possible from each other.

No consideration is given to other costs, such as foundations, electrical cables or installation and maintenance logistics.

Energy production ♥♥♥♥♥

Cost of support structures ♥♥♥♥♥

Cost of electrical infrastructure ♥♥♥♥♥

Cost of maintenance ♥♥♥♥♥



3 The foundations team

Shallower

Deeper

Rock

Only water depth and soil conditions influence the wind farm layout. Shallower waters reduce foundation

costs, and generally it's a good idea to avoid piling them into hard rocks.

Many turbines end up too close to one another,

increasing wake effects and reducing AEP considerably.

No consideration is given to the distance from the offshore electrical substations.

Rock

Energy production ♥♥♥♥♥

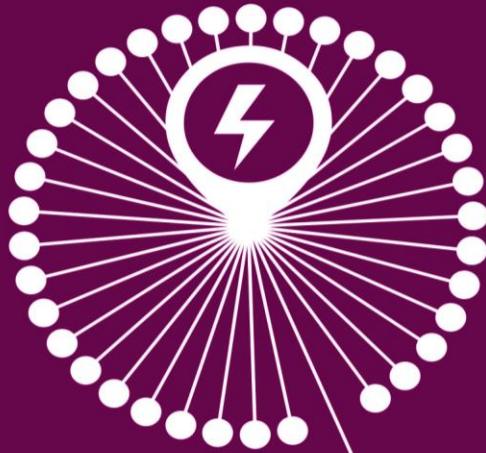
Cost of support structures ♥♥♥♥♥

Cost of electrical infrastructure ♥♥♥♥♥

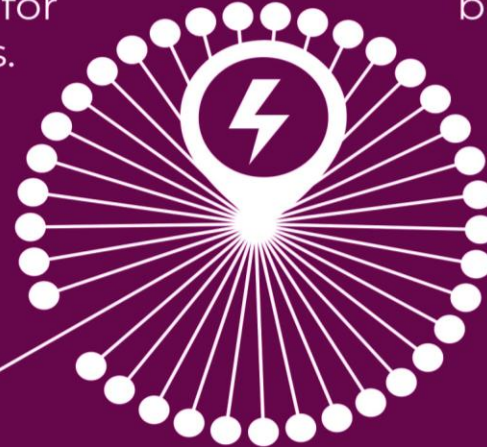
Cost of maintenance ♥♥♥♥♥



4 The electrical inter-array team



Turbines are placed as close as possible around the offshore substations to minimise electrical cable lengths and losses. AEP is hit pretty bad and there is no consideration whatsoever for bathymetry or soil conditions.



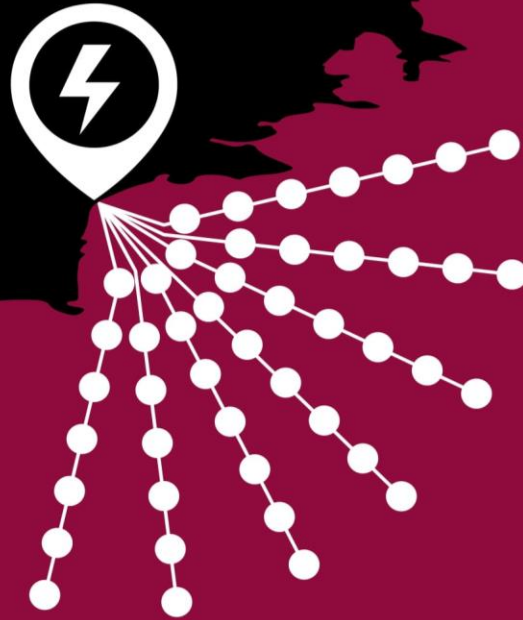
Energy production ♥♥♥♥♥

Cost of support structures ♥♥♥♥♥

Cost of inter-array cables ♥♥♥♥♥



5 The electrical export team



If this team was in charge of designing the layout, they would remove the offshore substations to save all the costs, and put the turbines as close as possible to the onshore electrical substation. Not only do other disciplines pay a big price for this absurd design, but also it would most likely not even be allowed by the authority.

Energy production ♥♥♥♥♥

Cost of support structures ♥♥♥♥♥

Cost of electrical export system ♥♥♥♥♥



6 The O&M team

Energy production ♥♥♥♥♥

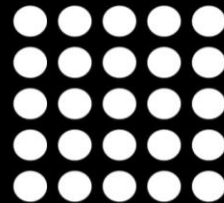
Cost of support structures ♥♥♥♥♥

Cost of maintenance ♥♥♥♥♥

If turbines must be placed offshore, then these would be in a nice tight grid close to shore.

●●●●● AEP is extremely low, and who knows what the seabed looks like right here.

If they could, this team would undoubtedly install the turbines onshore!



7 The commercial team

No wind farm, no costs!

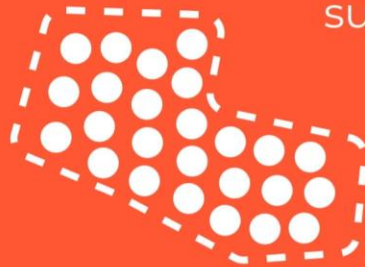
0 €/MWh

... and no environmental impact either!



8 The commercial team

More seriously though, one thing they would want to do is make the lease area smaller to reduce fees. Wake effects would be tremendously high, and AEP would suffer perhaps unnecessarily.



Energy production ♥♥♥♥♥

Lease fees ♥♥♥♥♥



9 All teams

No single discipline dominates this design, and instead is achieved with all teams working together.

No more barriers, no more competing objectives.

Most disciplines make a small sacrifice, but the entire system has the highest performance.

Everyone is happy.



Cost of electrical export system ♥♥♥♥♥

Cost of inter-array cables ♥♥♥♥♥

Cost of support structures ♥♥♥♥♥

Cost of maintenance ♥♥♥♥♥

Energy production ♥♥♥♥♥

Lease fees ♥♥♥♥♥

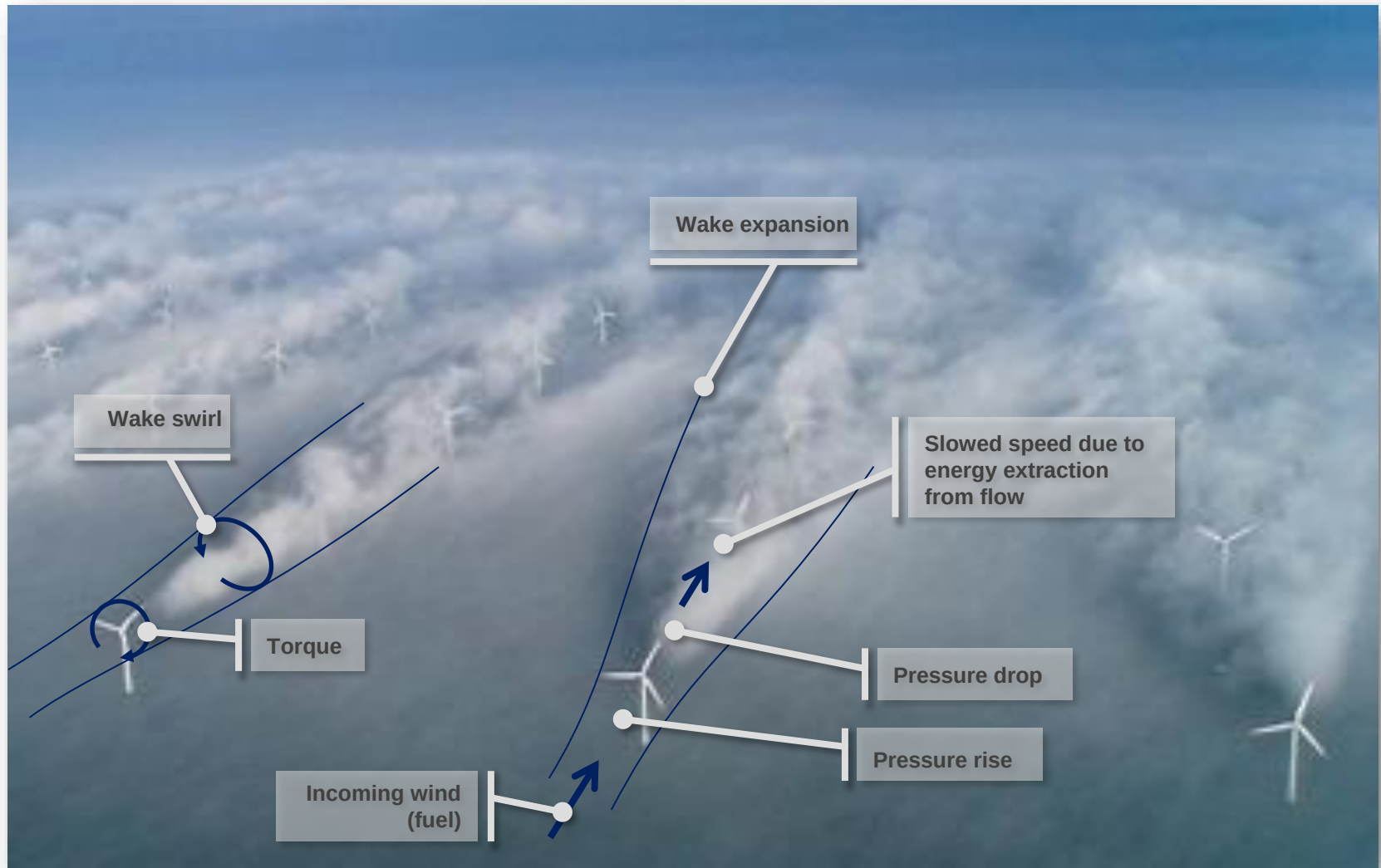


So, how do I find the optimal design?

Multidisciplinary Design Analysis and Optimisation

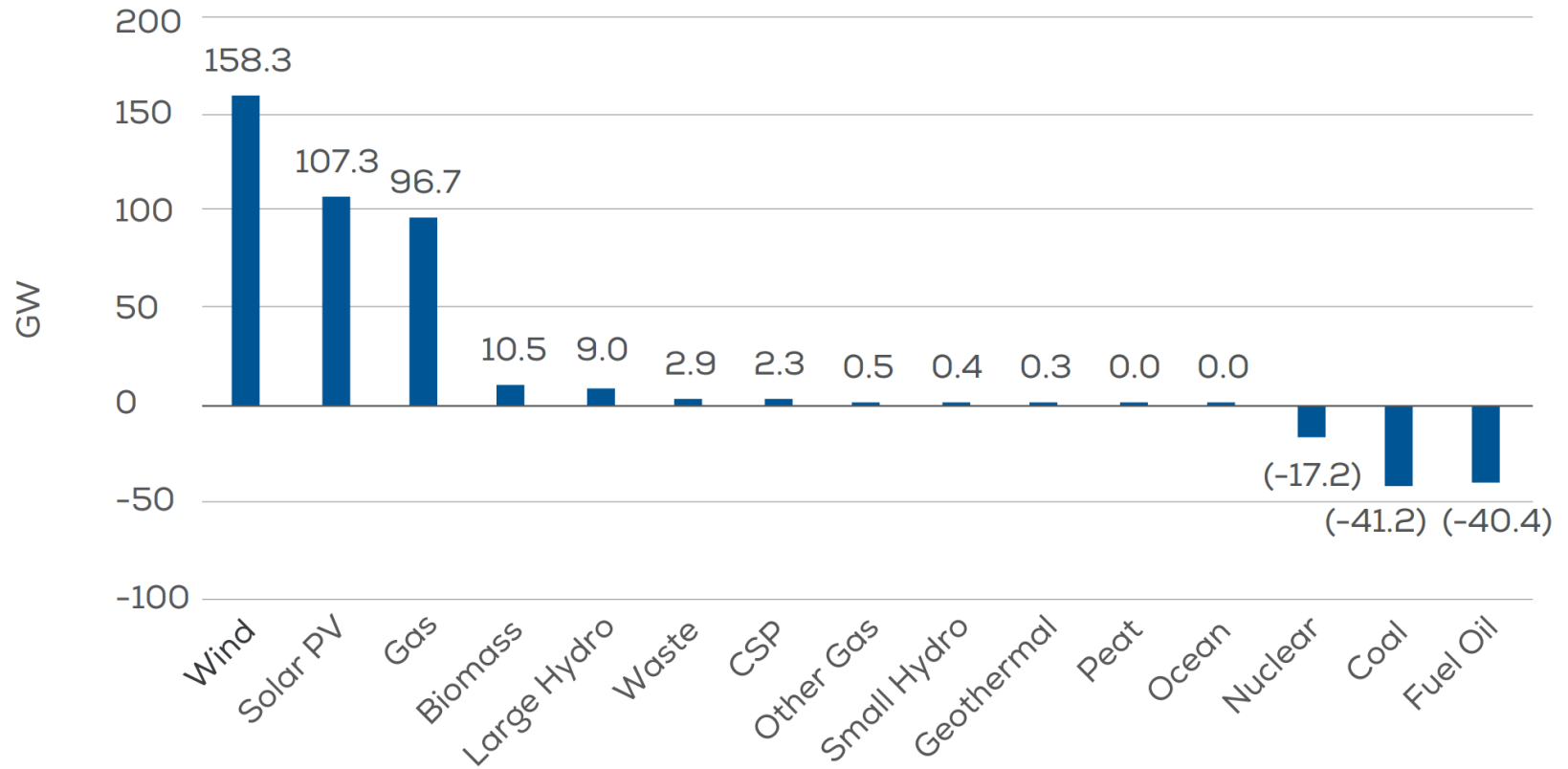


Making Torque from Wind



The Changing EU Power Mix

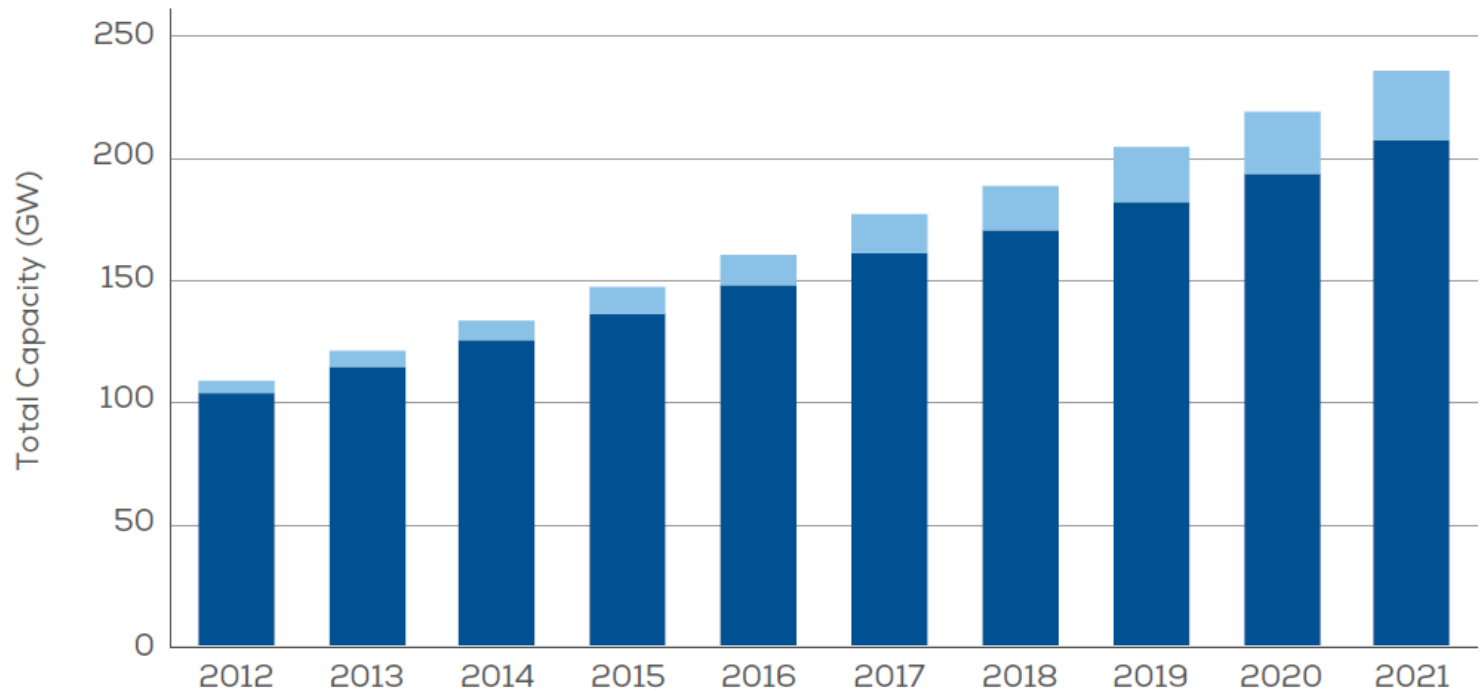
Net electricity installations in the EU from 2000 to 2017



Source: WindEurope

Wind Growth Trend in the EU

The growth of total wind energy capacity in Europe, 2012-21



Offshore	5	7	8	11	13	16	18	22	25	28
Onshore	104	114	126	136	148	161	170	182	194	207
Total	109	121	134	147	161	177	189	204	219	236

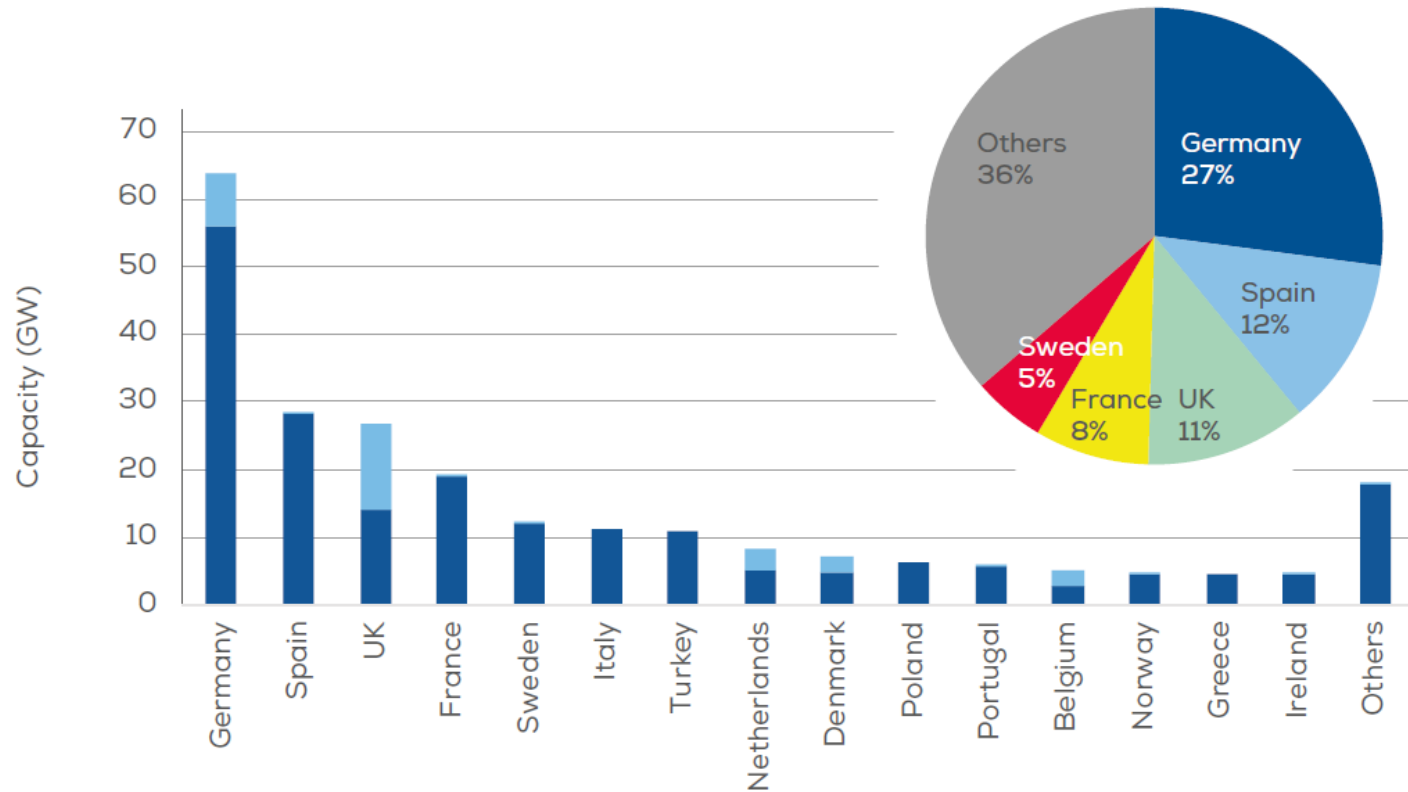
Source: WindEurope

Source: WindEurope



Shares by EU Country

Total installed wind power capacity by country



Offshore	8	0	12.7	0	0	-	-	3	2	-	0	2	0	-	0.03	0
Onshore	56	28	14	19	12	11	11	5	5	6	6	3	5	4	4	18
Total	64	28	26.7	19	12	11	11	8	7	6	6	5	5	4	4	18

(2019 data, source: WindEurope)



Capacity factors: fraction of time a wind plant operates at full power

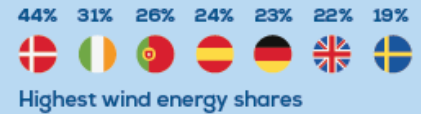
<https://www.smard.de/home>

Temporal variability on slow and fast time scales (discussed later in the wind resource chapter)

EUROPEAN WIND ENERGY GENERATION | 2021

15%

of Europe's electricity demand



207 GW
onshore wind capacity

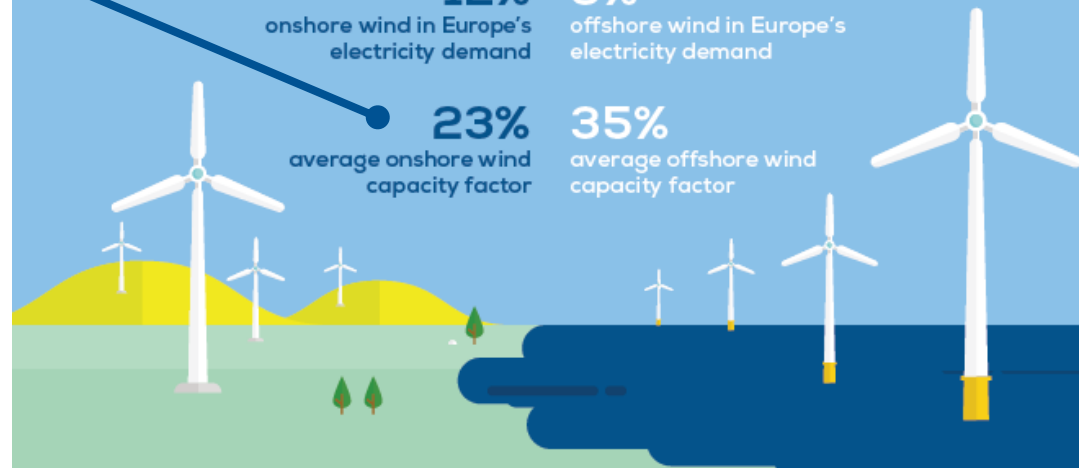
28 GW
offshore wind capacity

12%
onshore wind in Europe's electricity demand

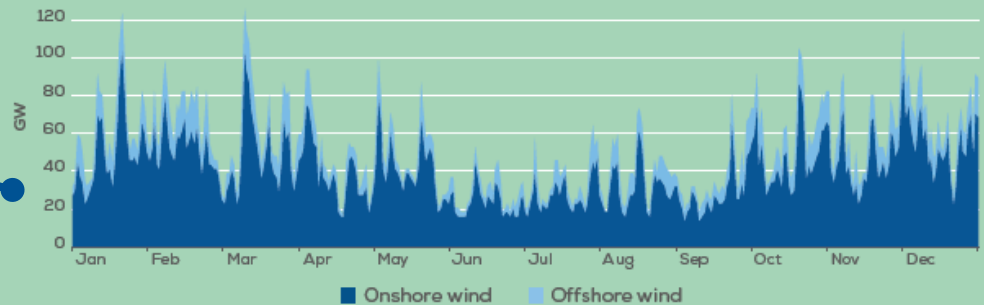
3%
offshore wind in Europe's electricity demand

23%
average onshore wind capacity factor

35%
average offshore wind capacity factor



European wind energy generation in 2021

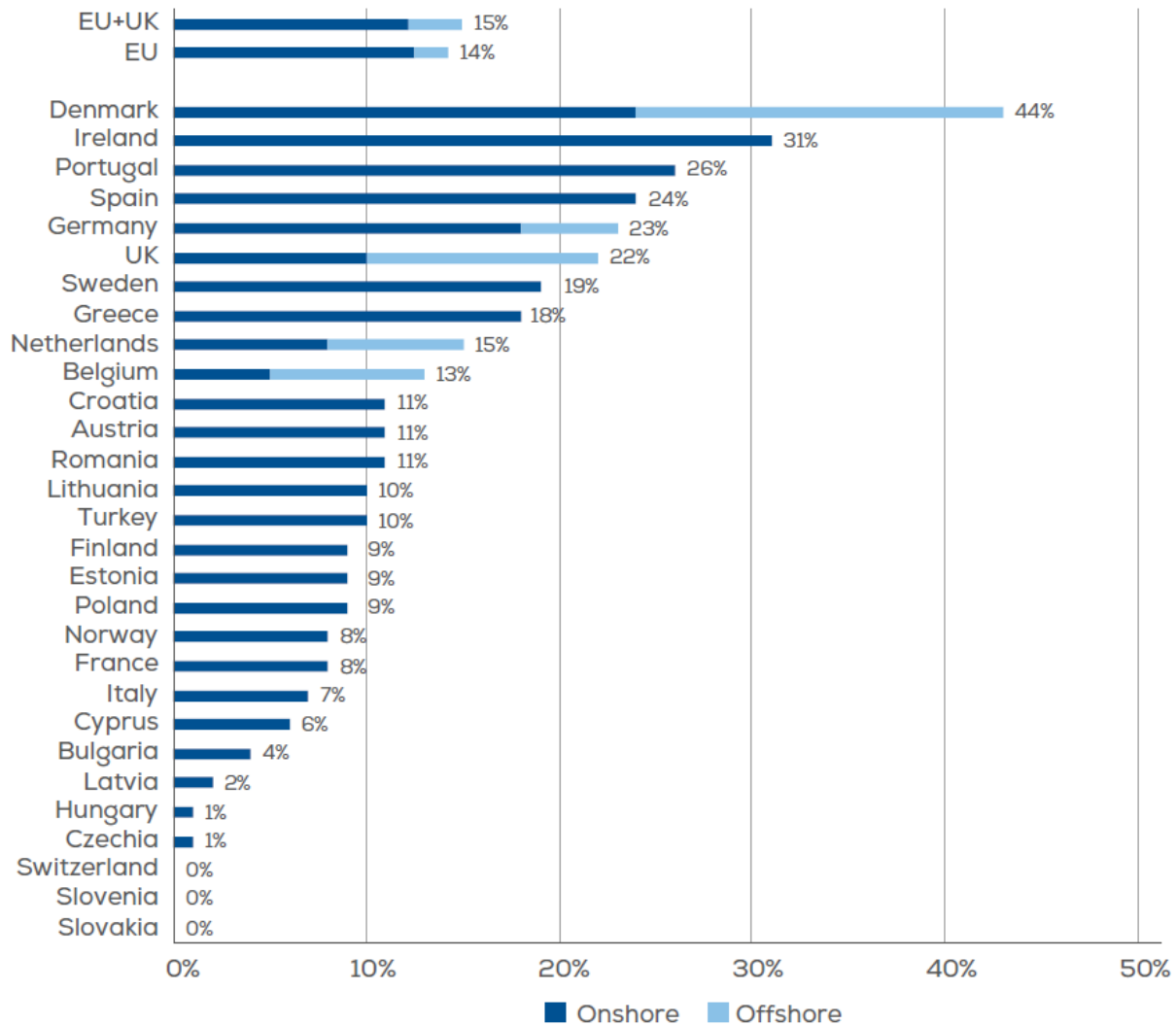


Data refers to EU Member States and the UK.



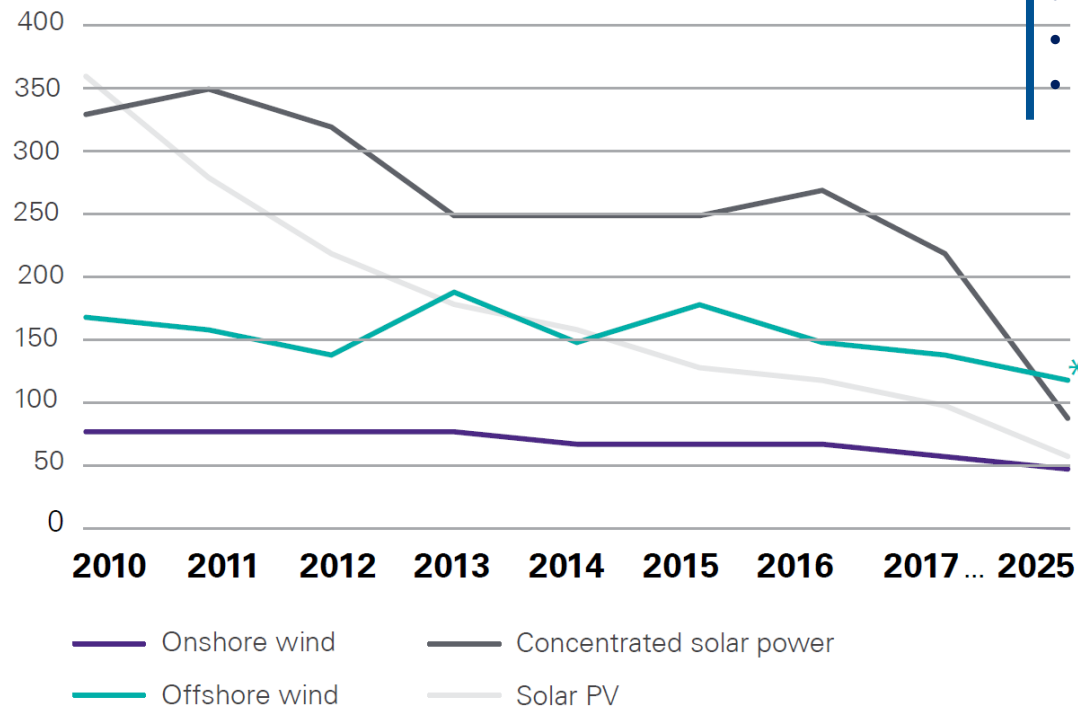
Wind Penetration Rates

Percentage of the average annual electricity demand covered by wind in 2021¹⁴



Cost of Wind Energy

LCOE of renewables and fossil fuel-based technologies
USD/MWh



LCOE: Levelized Cost of Energy

Cost reduction drivers:

- Technological improvement
- Economies of scale
- Competition
- Experience



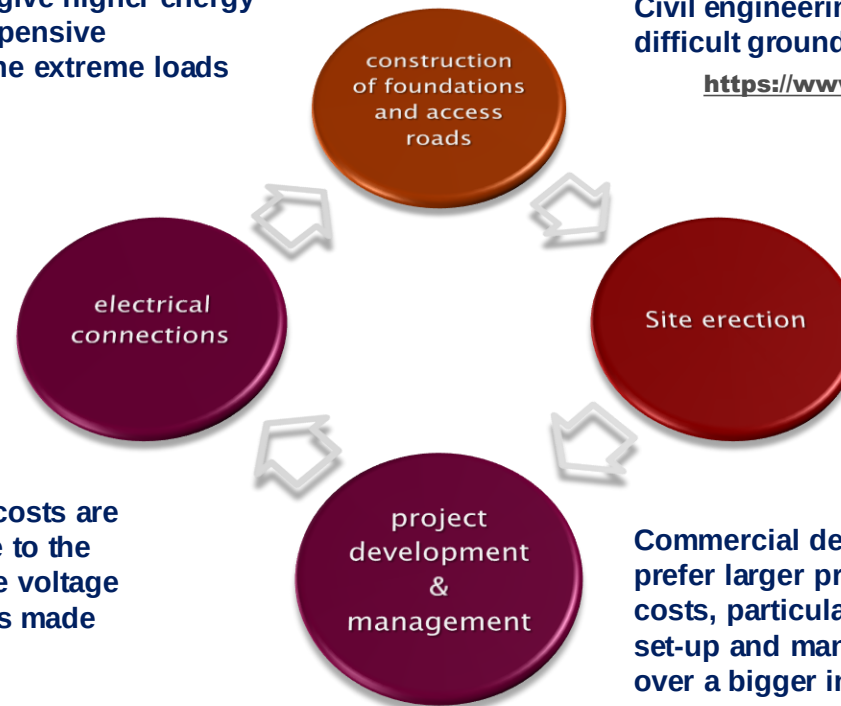
Wind Turbine Installation

For any wind turbine installation, there are certain additional activities:

High wind speed sites will give higher energy output but require more expensive foundations to withstand the extreme loads

Civil engineering costs are increased by difficult ground conditions

https://www.youtube.com/watch?v=fl6BMVw_B-Y

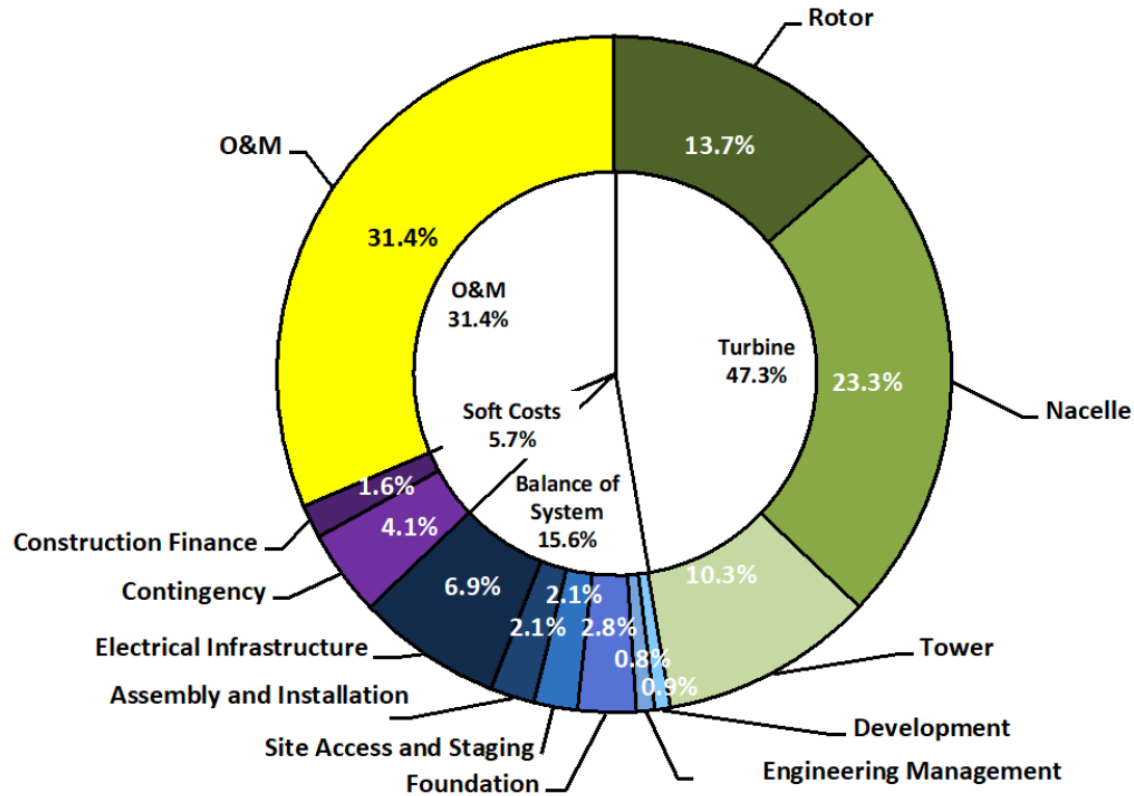


Electrical grid connection costs are determined by the distance to the point of connection and the voltage level at which connection is made

Commercial developers of wind farms will often prefer larger projects as, in that way, the fixed-costs, particularly grid connection and project set-up and management costs, may be spread over a bigger investment.

For flat onshore sites, which might be found typically in Denmark or North Germany, the total investment cost is approximately **1.3 times** the ex-works turbine cost.

Wind Turbine Component - LCOE Contribution

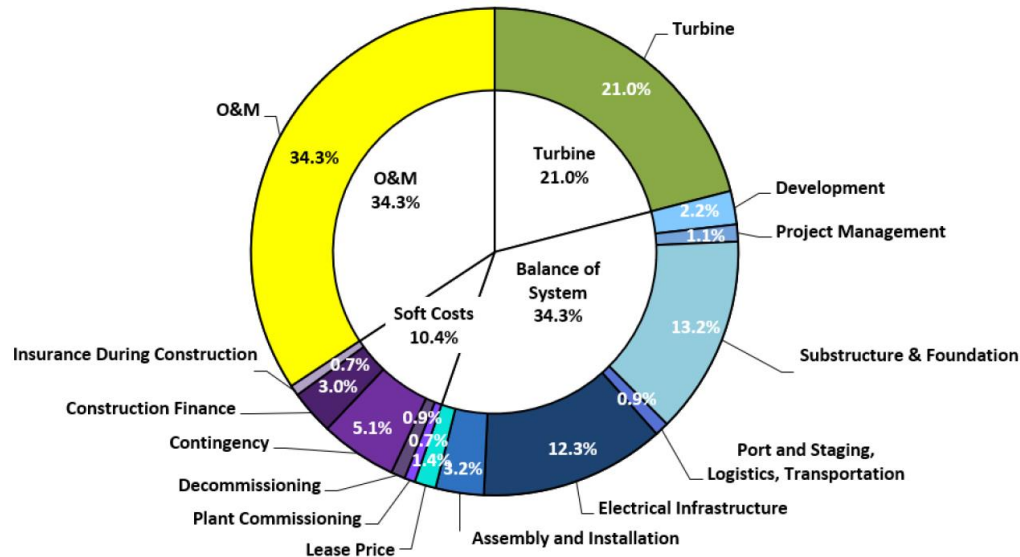


Component-level LCOE contribution for the 2019 land-based-wind reference project operating for 25 years

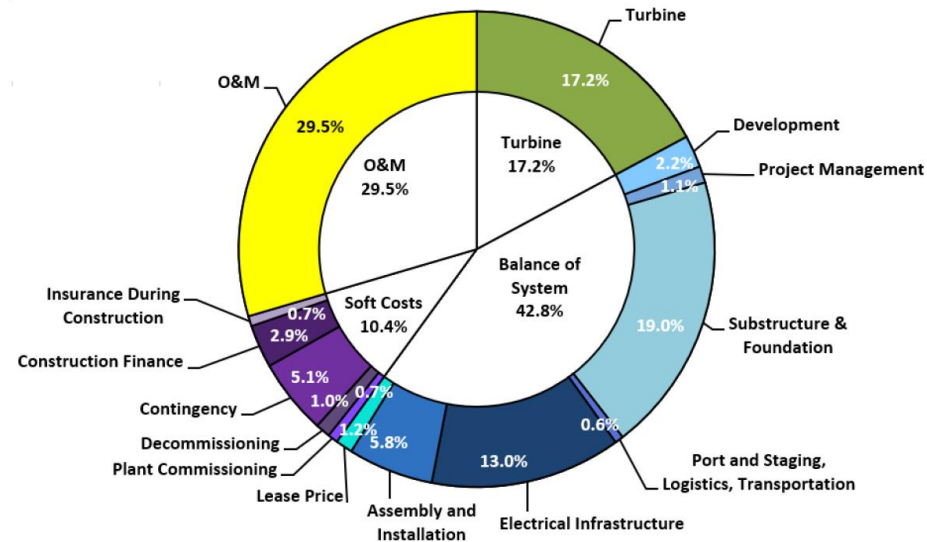
Source: NREL



Wind Farm Component-level LCOE Contribution



Component-level LCOE contribution for the 2019 floating offshore wind reference project operating for 25 years



Source: NREL



Issues Affecting Growth

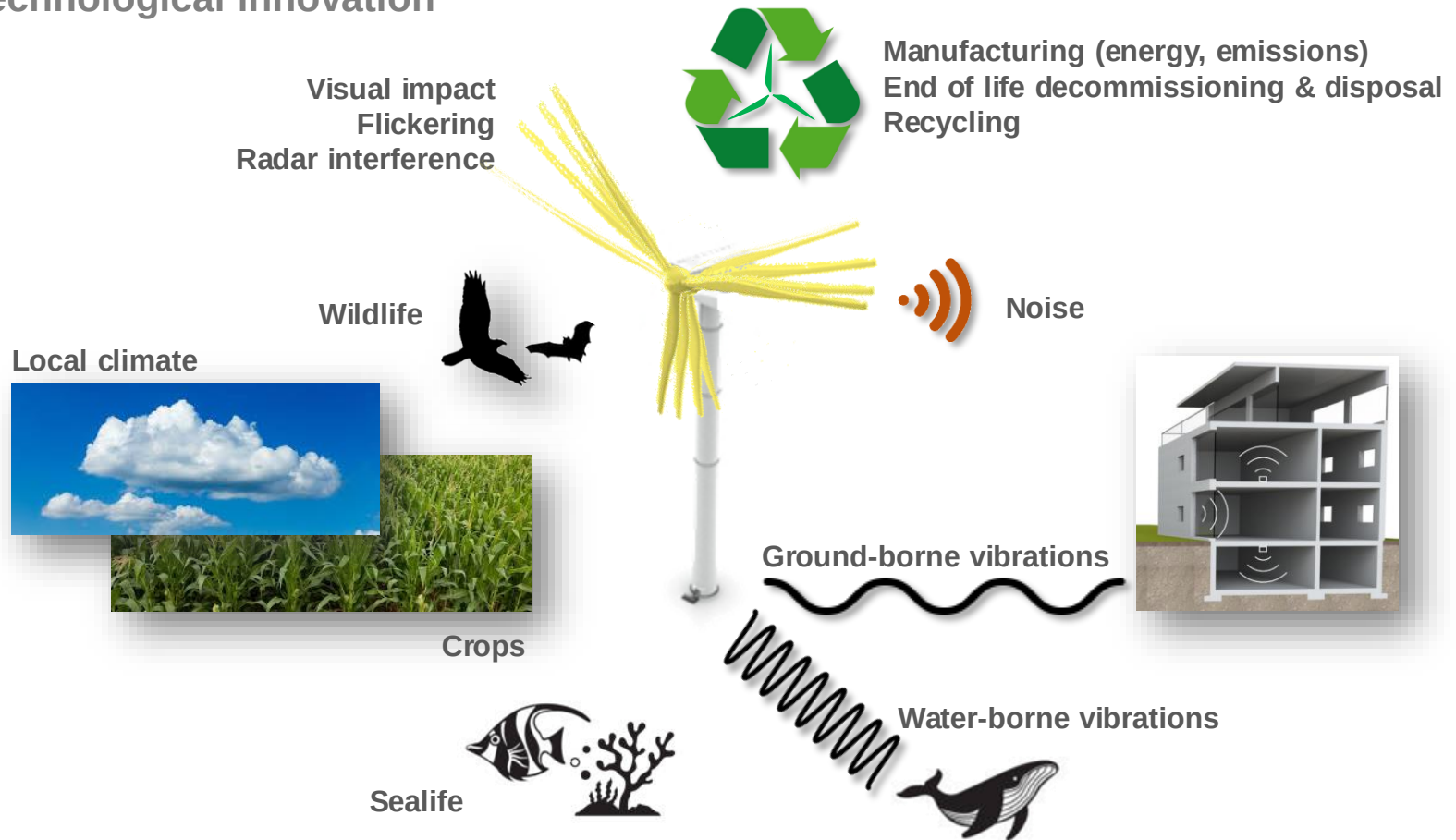
- Policy uncertainties
- Permitting & approval time/delays
- Grid capacity and transmission limits
- Skilled industrial/labor availability
- Cost of energy from wind
- Environmental impacts (wildlife, noise, visual, radar, ...)
- Social acceptance
- Availability of suitable on-shore sites
- Current technological limits (e.g.: for exploitation of deep-water off-shore resources)
- ...

Beyond reach of wind power technology

Affected by technological innovation

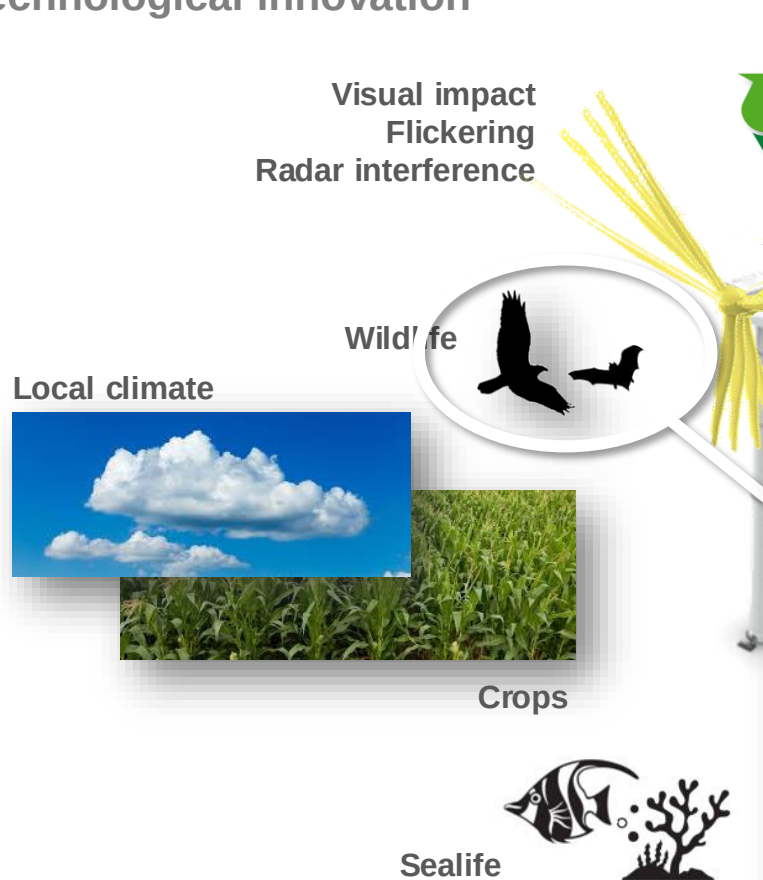
Wind Energy Impacts

Most social, economic and environmental impacts of wind energy are **positive**, but there are also some **negative** impacts, which can in part be mitigated by technological innovation



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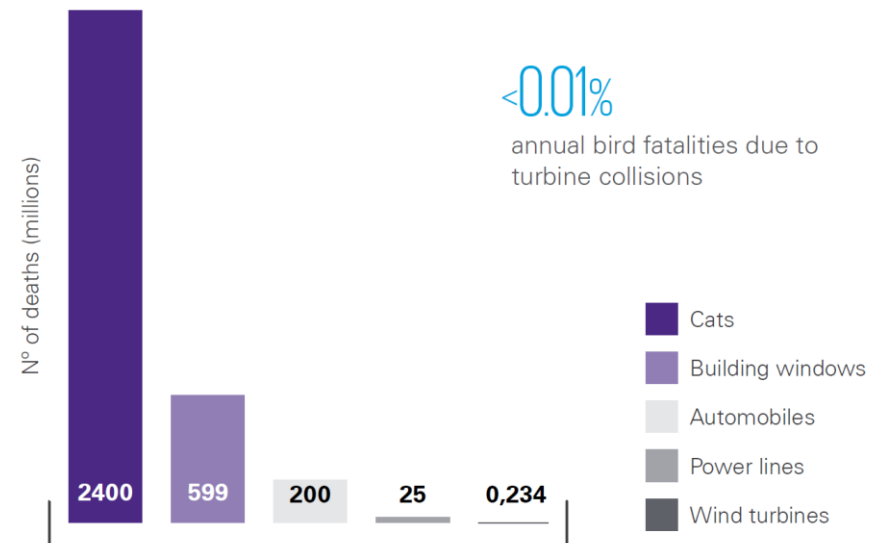


Manufacturing (energy, emissions)
End of life decommissioning & disposal
Recycling

Example: bird & bat mortality

Mitigating actions: choice of site, operation (curtailment, shut-down), deterrence

Global mortality of birds from human-related causes



USA. Direct Mortality of Birds from Anthropogenic Causes. Loss SR, Will T, Marra PP

(Source: KPMG Oct. 2019)

